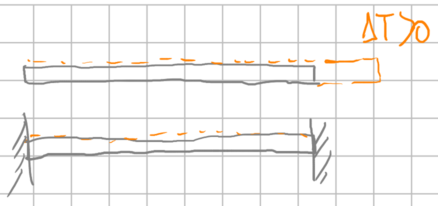
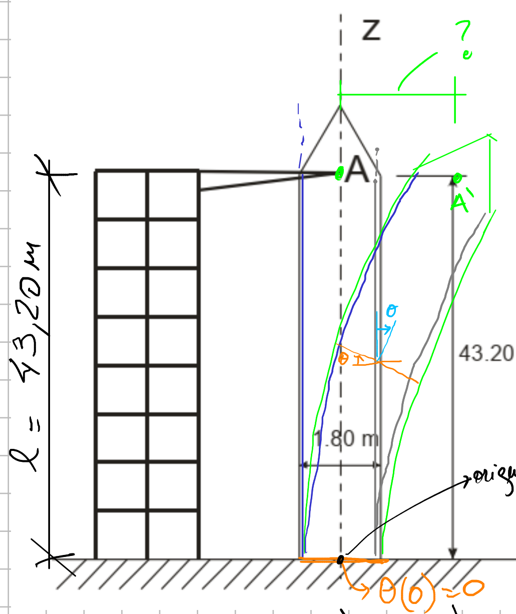


Pequeñas deformaciones

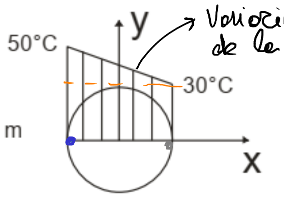
$\alpha = 10^{-5}\ \frac{cm}{cm\ °C}$

$\varepsilon_{ij}\ \alpha\Delta T = 20°C . 1\times10^{-5}\ \frac{cm}{cm\ °C} =$

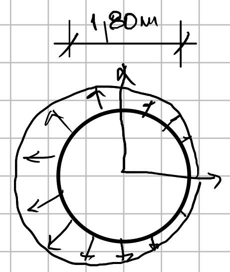
$= 2\times10^{-4}\ \frac{cm}{cm}$



$u_x(0) = 0$
$u_y(0) = 0$
$u_z(0) = 0$
(empotrado)



Variación lineal de la temperatura dentro del cuerpo



con respecto a una temperatura de referencia

$\underline{\underline{\varepsilon}} = \alpha\ \Delta T . \underline{\underline{I}}$ $\qquad \varepsilon_{ij} = \alpha\ \Delta T\ \delta_{ij}$

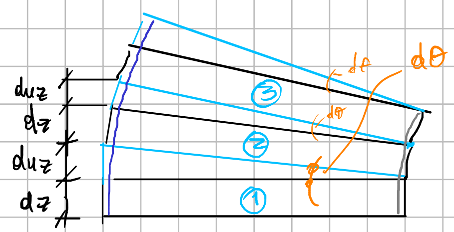
Dada deformación cuál es el desplazamiento

$\int \varepsilon_{xx} = \int \frac{\partial u_x}{\partial x} \Rightarrow u_x = f(x,y,z) + C(y,z)$

$\int \varepsilon_{zz} = \int \frac{\partial u_z}{\partial z}$

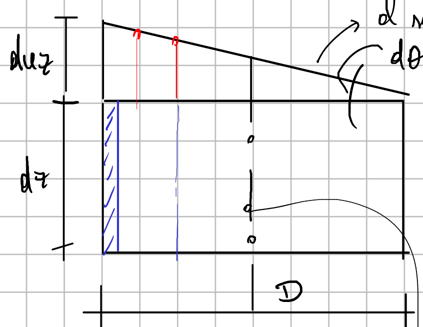
$u_x = \alpha\ \Delta T\ (x - x_0) + C(y,z)$

$\frac{\alpha\Delta T}{D} . \frac{z^2}{2}$



$\varepsilon_{zz} = \frac{du_z}{dz} = \left(\alpha\Delta T\ \underline{\underline{I}}\right)_{zz} = \alpha\ \Delta T . 1 \Rightarrow du_z = \alpha\ \Delta T\ dz$

al ser lineal la variación del incremento de temperatura



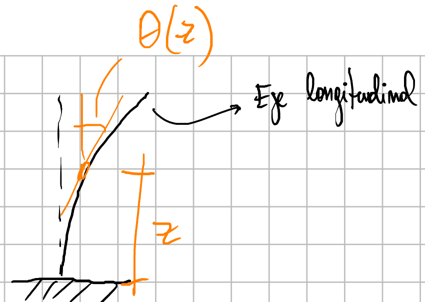
pequeñas deformaciones

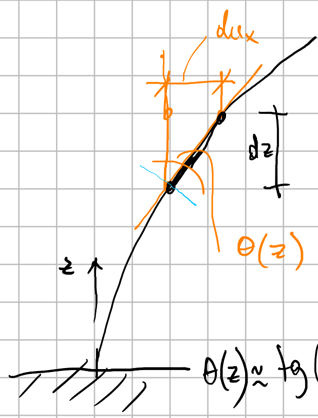
$tg(d\theta) \approx d\theta = \frac{du_z}{D} = \frac{\alpha\ \Delta T\ dz}{D}$

$\int_0^z d\theta = \int_0^z \frac{\alpha\ \Delta T\ dz}{D}$

$\theta(z) - \theta(0) = \frac{\alpha\ \Delta T}{D} \int_0^z dz = \frac{\alpha\ \Delta T}{D}(z - 0) = \frac{\alpha\ \Delta T}{D}\ z$

$\theta(z) = \frac{\alpha\ \Delta T}{D} . z$





$$\theta(z) \approx tg(\theta(z)) = \frac{du_x}{dz}$$

$$du_x = \theta(z)\cdot dz \Rightarrow \int du_x = \int \theta(z)\cdot dz$$

$$u_x = \int \theta(z)\cdot dz + C(x,y)$$

$$\underbrace{\int_0^z du_x}_{u_x(z) - u_x(0)} = \int_0^z \theta(z)\cdot dz = \int_0^z \frac{\alpha \Delta T}{D}\cdot z = \frac{\alpha \Delta T}{D}\,\frac{z^2}{2}\Big|_0^z = \frac{\alpha \Delta T}{D}\,\frac{z^2}{2}$$

$$u_x(z) - \underbrace{u_x(0)}_{0} = \frac{\alpha \Delta T}{D}\cdot\frac{z^2}{2} \Rightarrow u_x(z) = \frac{\alpha \Delta T}{D}\cdot\frac{z^2}{2}$$

$$u_x(\ell) = 10^{-5}\,\frac{cm}{cm\,°C}\cdot\frac{20°C}{180\,cm}\cdot\frac{(\overbrace{4320\,cm}^{\ell})^2}{2} = 10{,}368\,cm$$